

NAMANGAN, Uzbekistan

WMO: 386110

Lat: 40.983N

Lon: 71.562E

Elev: 474

StdP: 95.76

Time Zone: 5.00 (E05)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-8.5	-6.1	-12.1	1.4	-6.7	-9.8	1.7	-4.7	6.8	2.3	5.0	2.0	1.4	340	0.423

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.4	37.2	20.8	36.0	20.5	34.8	20.2	23.0	33.1	22.1	32.8	21.3	32.2	2.4	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
19.5	15.1	29.4	18.3	14.0	28.5	17.3	13.1	27.8	70.8	33.2	67.3	32.8	64.4	32.2	32.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 7.1	5.4	4.3	-10.2	39.6	3.4	1.0	-12.7	40.3	-14.7	41.0	-16.6	41.5	-19.1	42.3		
(5)			-10.8	25.1	3.3	2.2	-13.2	26.6	-15.2	27.9	-17.1	29.1	-19.5	30.7	(4)	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	15.5	0.8	3.6	10.9	17.0	22.4	27.0	28.6	27.2	22.3	15.5	8.1	2.0		
	DBStd	10.43	3.69	4.50	4.37	4.22	3.30	2.74	2.27	2.47	3.04	3.94	4.13	3.36		
	HDD10.0	852	286	182	42	4	0	0	0	0	0	6	83	249		
	HDD18.3	2191	544	413	231	71	7	0	0	0	6	103	308	507		
	CDD10.0	2859	0	3	71	214	384	508	576	532	368	177	25	0		
	CDD18.3	1156	0	0	1	32	132	258	318	274	124	16	0	0		
	CDH23.3	13020	0	0	12	271	1291	3116	4040	3082	1086	121	1	0		
	CDH26.7	6458	0	0	1	69	511	1646	2229	1580	402	21	0	0		
(14) Wind	WSAvg	2.1	1.5	1.9	2.2	2.4	2.5	2.6	2.3	2.2	2.1	2.0	1.7	1.5		
(15) Precipitation	PrecAvg	203	18	25	28	21	23	11	5	4	3	15	23	28		
	PrecMax	364	63	73	66	81	93	42	13	15	9	52	88	79		
	PrecMin	100	0	2	3	0	2	0	1	0	0	0	0	2		
	PrecStd	60	13	17	15	15	20	11	4	3	2	13	23	18		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	11.0	17.1	25.1	31.2	34.9	38.8	39.2	37.6	33.7	29.1	22.2	12.2		
		MCWB	6.5	9.3	13.5	16.9	18.3	20.3	21.2	21.5	19.8	17.1	13.3	7.5		
	2%	DB	8.9	13.8	22.2	28.8	33.1	36.9	37.8	35.9	32.1	26.3	18.7	9.9		
		MCWB	5.4	7.8	12.6	16.6	18.1	20.0	21.0	21.0	18.8	15.9	11.6	6.2		
	5%	DB	7.1	11.6	20.0	26.5	31.4	35.2	36.2	34.9	30.8	24.3	16.1	8.1		
		MCWB	4.3	6.7	11.9	15.7	17.7	19.6	20.8	20.9	18.2	14.5	10.7	5.1		
	10%	DB	5.7	9.7	17.9	24.2	29.8	33.9	35.0	33.5	29.1	22.3	14.1	6.8		
		MCWB	3.4	5.8	11.1	14.9	17.3	19.3	20.4	20.4	17.7	13.7	9.6	4.4		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	7.1	9.9	14.7	19.3	20.4	23.1	25.8	24.0	21.4	18.0	14.1	8.6		
		MCDB	10.1	15.9	21.8	27.7	31.3	34.5	35.0	33.7	30.8	26.3	20.2	11.1		
	2%	WB	5.7	8.4	13.6	17.9	19.4	21.7	23.3	22.8	20.1	16.3	12.4	6.7		
		MCDB	8.1	12.6	20.3	25.6	30.0	33.1	33.5	32.5	29.7	24.5	17.1	8.8		
	5%	WB	4.7	7.3	12.6	16.8	18.7	21.0	22.2	22.1	19.0	15.4	11.2	5.5		
		MCDB	6.6	10.8	18.7	24.0	28.7	32.6	33.3	32.3	28.8	22.9	15.3	7.7		
	10%	WB	3.6	6.2	11.6	15.8	18.0	20.2	21.4	21.3	18.1	14.4	10.0	4.4		
		MCDB	5.4	9.2	17.0	22.6	27.3	31.4	32.6	31.5	27.7	20.8	13.7	6.4		
(35) Mean Daily Temperature Range	5% DB	MDBR	6.3	7.4	9.2	10.5	11.9	12.7	12.4	12.2	12.4	11.1	8.5	6.1		
		MCDBR	8.0	10.3	12.5	13.5	14.2	14.2	13.6	13.2	13.5	13.0	11.3	8.6		
	5% WB	MCWBR	5.6	6.2	6.4	6.0	5.2	5.1	5.4	5.7	6.1	6.1	6.4	5.9		
		MCWBR	7.0	9.1	11.0	11.6	12.5	12.9	12.7	12.5	12.9	11.6	9.7	7.6		
(40) Clear-Sky Solar Irradiance	taub		0.361	0.412	0.488	0.508	0.494	0.493	0.476	0.446	0.437	0.478	0.434	0.363		
	taud		2.179	2.047	1.883	1.870	1.962	1.979	2.023	2.102	2.060	1.898	1.977	2.153		
	Ebn at Noon		760	770	750	770	794	794	803	817	791	684	654	717		
	Edh at Noon		102	136	179	194	181	178	169	152	148	155	125	97		
(44) All-Sky Solar Radiation	RadAvg		2.03	2.87	3.92	5.25	6.38	7.28	7.42	6.73	5.49	3.56	2.25	1.72		
	RadStd		0.18	0.32	0.33	0.45	0.41	0.32	0.27	0.30	0.20	0.37	0.30	0.22		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	+0.35	N/A	N/A	+0.62	-0.65	-1.01	N/A	N/A	+55	+49
Regional (9 neighbors)	N/A	N/A	N/A	+0.51	N/A	N/A	N/A	N/A	+96	+64

Nomenclature: See separate page